

Module 5 — Newtonian World and Astrophysics

This module covers thermal physics, circular motion, oscillations, gravitational fields, and astrophysics — the Year 2 core content.

5.1 Thermal Physics

Internal Energy

Internal energy: The sum of the randomly distributed kinetic energies and potential energies of all the molecules in a system. Internal energy increases when temperature rises or during a change of state.

- Kinetic energy of molecules is related to temperature
- Potential energy of molecules is related to the intermolecular forces (bonds)
- During a change of state, temperature remains constant — all energy input goes into breaking or forming intermolecular bonds (changing potential energy, not kinetic)

Specific Heat Capacity

Specific heat capacity (c): The energy required to raise the temperature of 1 kg of a substance by 1 K (or 1 °C).

$$Q = mc\Delta\theta$$

Where Q = energy (J), m = mass (kg), c = specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$), $\Delta\theta$ = change in temperature (K or °C).

Specific Latent Heat

Specific latent heat (L): The energy required to change the state of 1 kg of a substance without changing its temperature.

Specific latent heat of fusion (L_f): For solid $\leftrightarrow$ liquid transitions.

Specific latent heat of vaporisation (L_v): For liquid $\leftrightarrow$ gas transitions.

$$Q = mL$$

Exam Tip: In heating/cooling curve questions, the flat sections represent changes of state where $Q = mL$ applies. The sloped sections represent temperature changes where $Q = mc\Delta\theta$ applies. L_v is always greater than L_f for the same substance because more bonds are broken.

The Ideal Gas

Ideal gas assumptions: (1) Molecules are point particles with negligible volume. (2) No intermolecular forces except during collisions. (3) Collisions are perfectly elastic. (4) Collision time is negligible compared to time between collisions. (5) Molecules move randomly with a range of speeds.

Gas Laws

Law	Relationship	Condition
Boyle's Law	$p \propto 1/V$ ($pV = \text{constant}$)	Constant temperature and mass
Charles's Law	$V \propto T$ ($V/T = \text{constant}$)	Constant pressure and mass
Gay-Lussac's Law	$p \propto T$ ($p/T = \text{constant}$)	Constant volume and mass

Temperature must always be in kelvin: $T(K) = T(^{\circ}C) + 273.15$

Ideal Gas Equation

$$pV = nRT = NkT$$

Where p = pressure (Pa), V = volume (m^3), n = number of moles, R = molar gas constant = $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, N = number of molecules, k = Boltzmann constant = $1.38 \times 10^{-23} \text{ J K}^{-1}$, T = absolute temperature (K).

Note: $R = N_A k$, where N_A = Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$.

Kinetic Theory and Molecular Kinetic Energy

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

Where m = mass of one molecule, $\langle c^2 \rangle$ = mean square speed of the molecules.

The mean kinetic energy of a molecule:

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

- The average kinetic energy of a gas molecule is proportional to the absolute temperature
- At a given temperature, all gas molecules have the same average kinetic energy (regardless of mass)
- Heavier molecules move more slowly on average than lighter ones at the same temperature
- The root mean square (rms) speed: $c_{\text{rms}} = \sqrt{\langle c^2 \rangle}$

Exam Tip: Be careful with units. Pressure must be in Pa, volume in m^3 , and temperature in K.

Remember: $1 \text{ litre} = 10^{-3} \text{ m}^3$, $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$.

5.2 Circular Motion

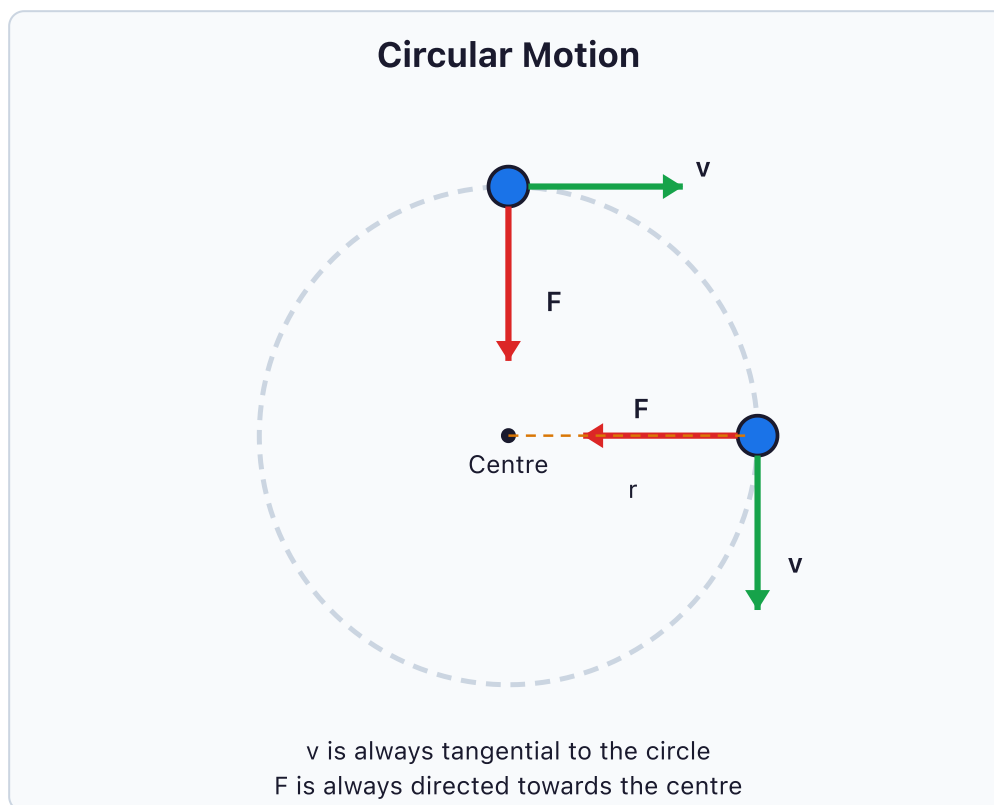
Angular velocity (ω): The rate of change of angle, measured in radians per second (rad s^{-1}).

$$\omega = 2\pi f = 2\pi / T$$

Where f = frequency (Hz), T = period (s).

Linear speed and angular velocity:

$$v = \omega r = 2\pi r / T$$



Centripetal Acceleration and Force

An object moving in a circle at constant speed is accelerating because its velocity (direction) is continuously changing. The acceleration is directed towards the centre of the circle.

$$a = v^2 / r = \omega^2 r$$

$$F = mv^2 / r = m\omega^2 r$$

- The centripetal force is not a new type of force — it is the resultant force directed towards the centre, provided by gravity, tension, friction, normal force, etc.
- The centripetal force does no work because it is always perpendicular to the velocity (displacement)

- If the centripetal force is removed, the object moves in a straight line tangent to the circle (Newton's First Law)

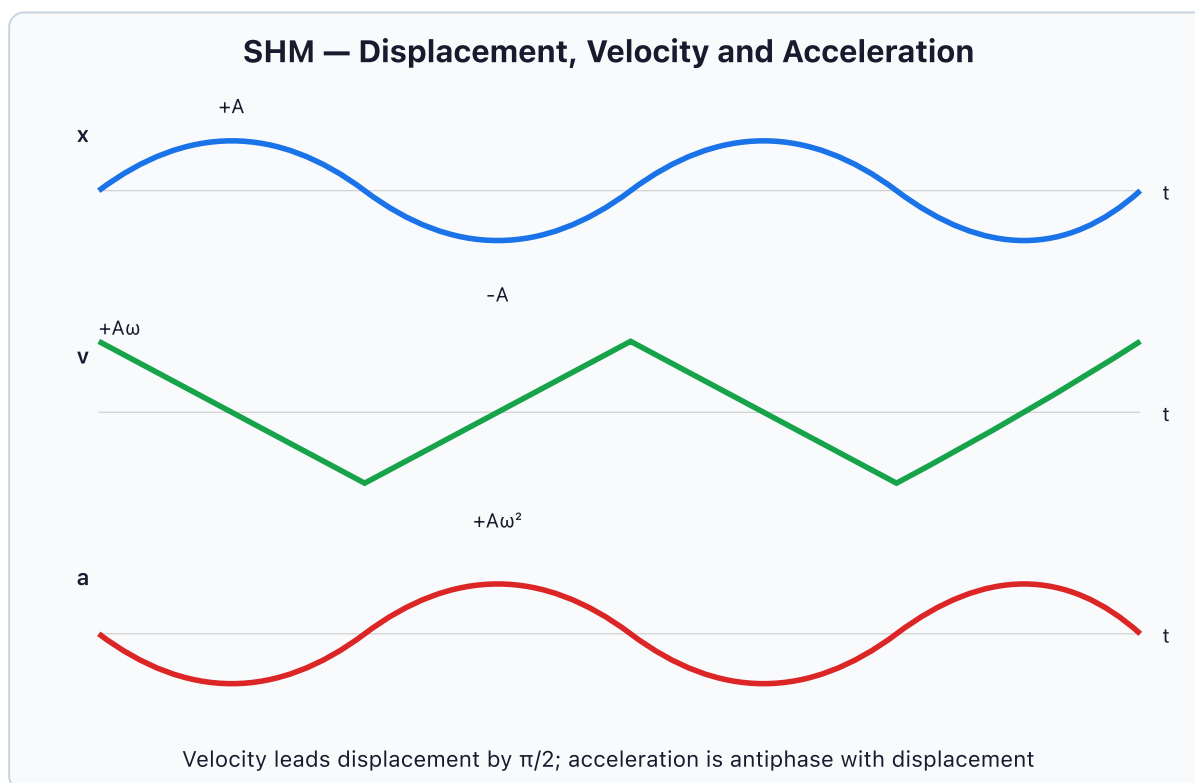
Exam Tip: Never say "centripetal force acts on the object" in isolation. Always state what provides the centripetal force (e.g., "gravity provides the centripetal force for orbital motion").

5.3 Simple Harmonic Motion (SHM)

Simple harmonic motion: An oscillation in which the acceleration is proportional to the displacement and always directed towards the equilibrium position.

$$a = -\omega^2 x$$

Where a = acceleration (m s^{-2}), ω = angular frequency (rad s^{-1}), x = displacement from equilibrium (m). The minus sign indicates acceleration is always directed towards equilibrium (restoring force).



SHM Equations

$$x = A \cos(\omega t) \quad \text{or} \quad x = A \sin(\omega t)$$

$$v = -A\omega \sin(\omega t) \quad \text{or} \quad v = A\omega \cos(\omega t)$$

$$v = \pm\omega\sqrt{(A^2 - x^2)}$$

Maximum speed: $v_{\max} = A\omega$ (at equilibrium, $x = 0$)

Maximum acceleration: $a_{\max} = A\omega^2$ (at extremes, $x = \pm A$)

Period of Oscillation

Mass-spring system:

$$T = 2\pi\sqrt{(m / k)}$$

Where m = mass (kg), k = spring constant (N m^{-1}). Period is independent of amplitude.

Simple pendulum:

$$T = 2\pi\sqrt{(L / g)}$$

Where L = length of pendulum (m), g = gravitational field strength (m s^{-2}). Valid for small angles ($< \sim 10^\circ$). Period is independent of mass and amplitude.

Energy in SHM

- At equilibrium: KE is maximum, PE is zero (or minimum)
- At extremes: KE is zero, PE is maximum
- Total energy = KE + PE = constant (for undamped SHM)
- $E_k = \frac{1}{2}m\omega^2(A^2 - x^2)$
- $E_{\text{total}} = \frac{1}{2}m\omega^2A^2$

Damping

Damping: The reduction in amplitude of an oscillation over time due to energy being transferred to the surroundings (e.g., by friction or air resistance).

- **Light damping:** Amplitude decreases gradually over many oscillations. Frequency is approximately unchanged.
- **Heavy damping:** Amplitude decreases rapidly. The system returns to equilibrium slowly without oscillating.
- **Critical damping:** The system returns to equilibrium in the shortest possible time without oscillating. Used in car suspension, door closers.

Forced Oscillations and Resonance

Resonance: Occurs when the driving frequency matches the natural frequency of a system. The amplitude of oscillation reaches a maximum and energy transfer from driver to oscillator is most efficient.

- At resonance, the amplitude is limited only by damping
- Increased damping reduces the resonance peak, broadens it, and slightly lowers the resonant frequency
- Examples: Tacoma Narrows bridge, tuning a radio, wine glass shattering, MRI machines

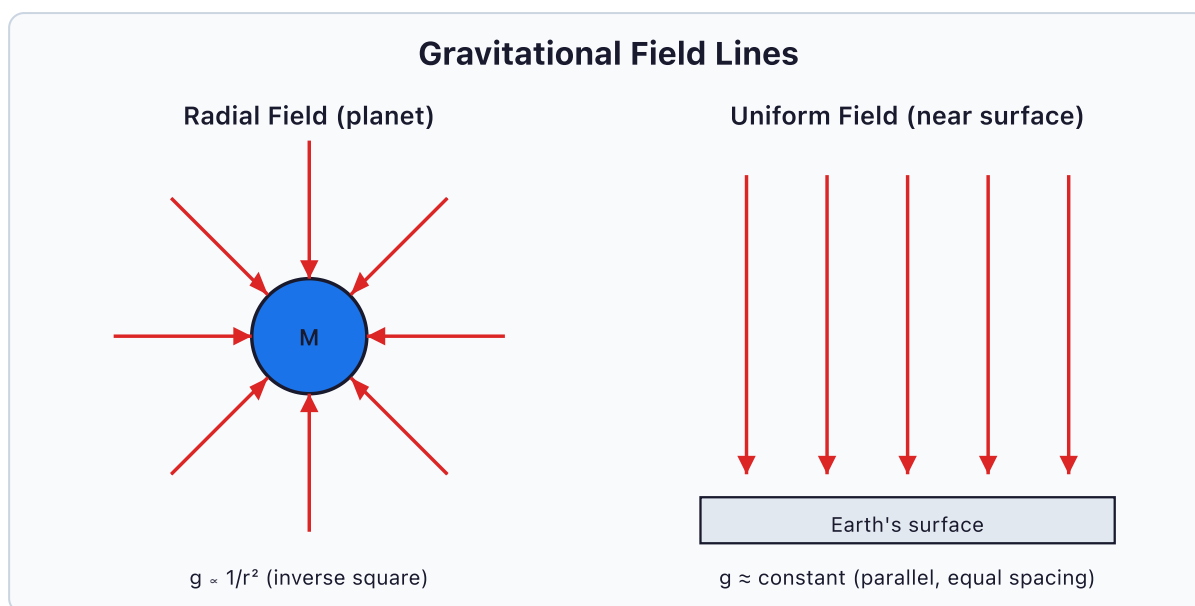
5.4 Gravitational Fields

Gravitational field: A region of space in which a mass experiences a gravitational force. All objects with mass create a gravitational field.

Newton's Law of Gravitation

$$F = -GMm / r^2$$

Where G = gravitational constant = $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, M and m are the two masses (kg), r = distance between their centres (m). The negative sign indicates an attractive force.



Gravitational Field Strength

$$g = F / m = GM / r^2$$

Units: N kg^{-1} or m s^{-2} . g is the force per unit mass and equals the acceleration due to gravity.

- On the surface of Earth: $g \approx 9.81 \text{ N kg}^{-1}$
- g follows an inverse square law with distance from the centre of the planet
- Inside a uniform sphere, g decreases linearly to zero at the centre

Gravitational Potential

Gravitational potential (V_g): The work done per unit mass in bringing a small test mass from infinity to that point.

$$V_g = -GM / r$$

Units: J kg^{-1} . Gravitational potential is always negative (since gravity is attractive). At infinity, $V_g = 0$.

Gravitational potential energy:

$$E_p = mV_g = -GMm / r$$

Exam Tip: The minus sign in $V_g = -GM/r$ is crucial. It means you need to do work (add energy) to move a mass further from the gravitating body. Do not confuse this with mgh , which is only valid near Earth's surface.

Orbital Motion and Kepler's Laws

Kepler's Three Laws:

1. **First Law:** Planets orbit in ellipses with the Sun at one focus
2. **Second Law:** A line joining a planet to the Sun sweeps out equal areas in equal times (planets move faster when closer to the Sun)
3. **Third Law:** $T^2 \propto r^3$. The square of the orbital period is proportional to the cube of the orbital radius.

Deriving Kepler's Third Law: For a circular orbit, gravity provides the centripetal force:

$$GMm / r^2 = mv^2 / r = m\omega^2 r = m(2\pi/T)^2 r$$

$$T^2 = (4\pi^2 / GM) r^3$$

Orbital speed:

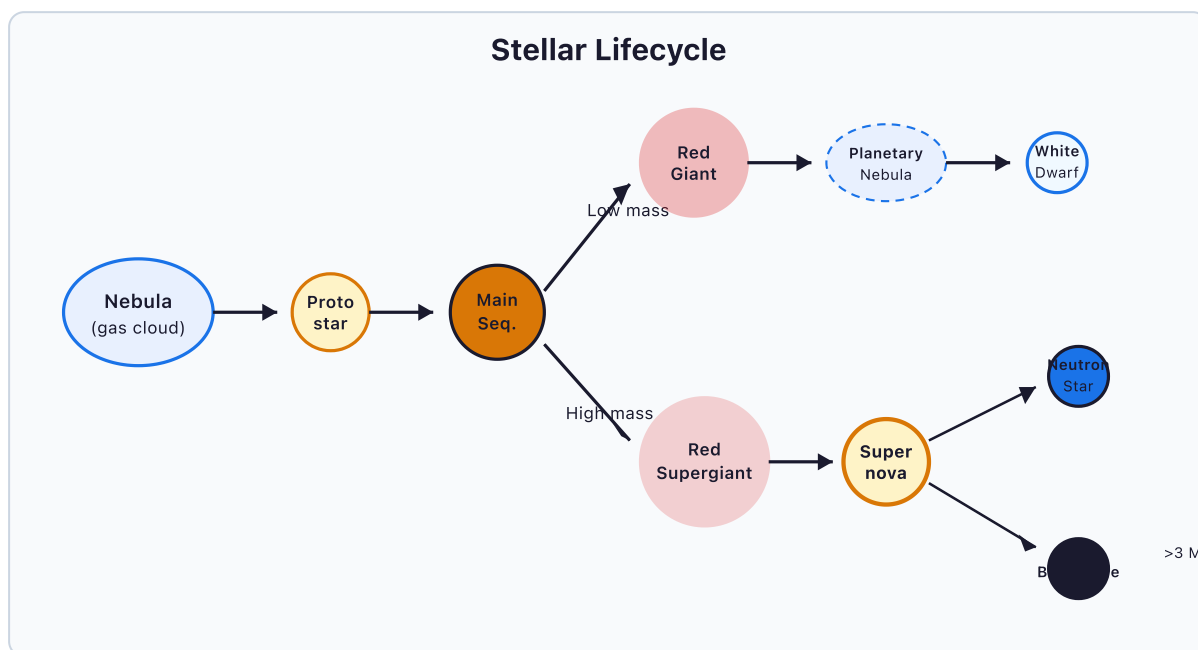
$$v = \sqrt{GM / r}$$

Geostationary Orbits

- Period = 24 hours (same as Earth's rotation)
- Orbits above the equator
- Same direction as Earth's rotation (west to east)
- Appears stationary relative to a point on Earth's surface
- Orbital radius $\approx 42,200$ km from Earth's centre (altitude $\sim 35,800$ km)
- Used for: communication satellites, weather satellites, TV broadcasting

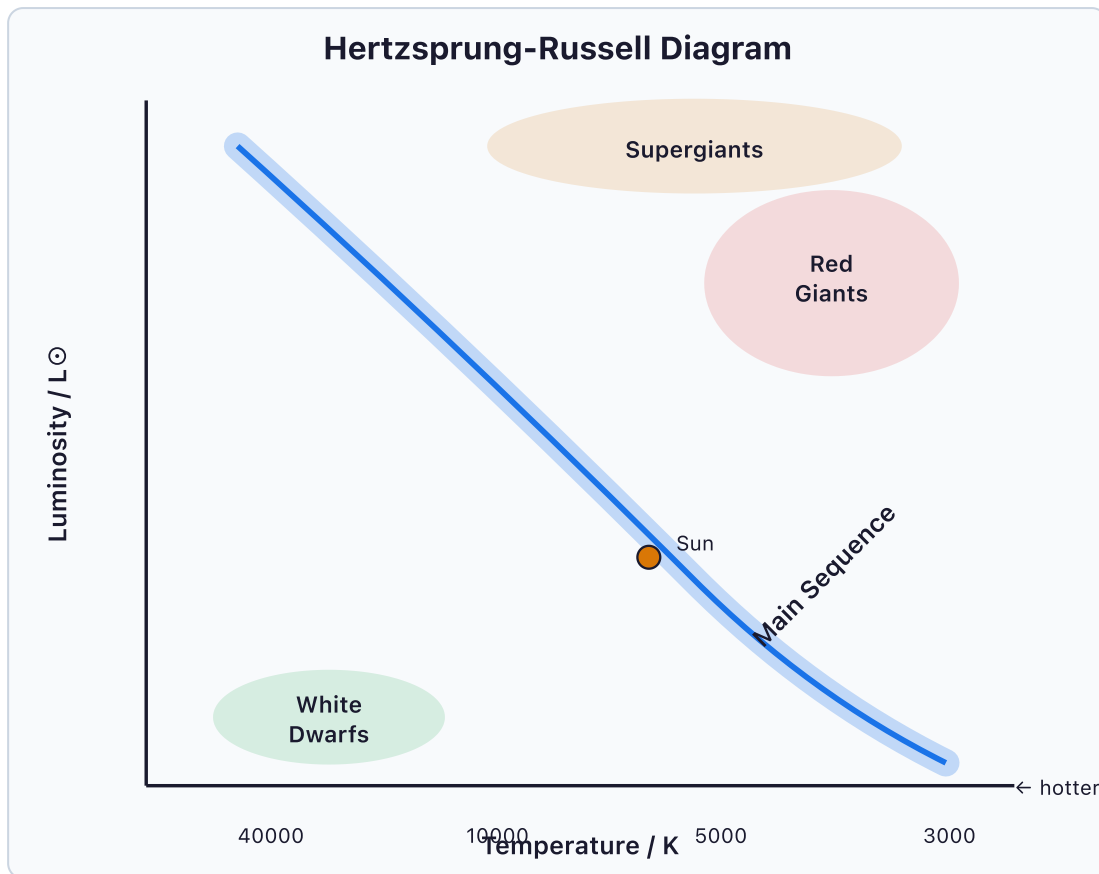
5.5 Astrophysics

Stellar Evolution



Stage	Low/Medium Mass Star (like our Sun)	High Mass Star
Birth	Nebula → protostar (gravitational collapse, temperature rises)	
Main sequence	Hydrogen fusion in the core. Equilibrium between radiation pressure (outward) and gravity (inward). Lasts millions to billions of years.	
After main sequence	Red giant (helium fusion begins, outer layers expand and cool)	Red supergiant (fuses heavier elements up to iron in shells)
Death	Planetary nebula → white dwarf (supported by electron degeneracy pressure)	Supernova explosion → neutron star (supported by neutron degeneracy) or black hole (if remnant > ~3 solar masses)

Hertzprung-Russell (HR) Diagram



- X-axis: temperature (decreasing left to right, or spectral class O B A F G K M)
- Y-axis: luminosity (increasing upward, often logarithmic)
- **Main sequence:** Diagonal band from top-left (hot, luminous) to bottom-right (cool, dim). Most stars (~90%).
- **Red giants / supergiants:** Top-right (cool but very luminous due to large size)
- **White dwarfs:** Bottom-left (hot but dim due to small size)

Wien's Displacement Law

$$\lambda_{\max} T = 2.898 \times 10^{-3} \text{ m K}$$

Where $\lambda_{\max}$ = peak wavelength of the emission spectrum (m), T = surface temperature (K). Hotter stars peak at shorter wavelengths (bluer); cooler stars peak at longer wavelengths (redder).

Stefan-Boltzmann Law

$$L = 4\pi r^2 \sigma T^4$$

Where L = luminosity (total power output, W), r = radius of the star (m), σ = Stefan-Boltzmann constant $= 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, T = surface temperature (K).

Exam Tip: To compare two stars, use ratios. For example: $L_1/L_2 = (r_1/r_2)^2 \times (T_1/T_2)^4$. This avoids needing to know σ or use very large numbers.

Hubble's Law and the Expanding Universe

Cosmological redshift: The increase in wavelength (decrease in frequency) of light from distant galaxies due to the expansion of space. The galaxy is not moving through space — space itself is expanding.

$$z = \Delta\lambda / \lambda = v / c$$

Where z = redshift, $\Delta\lambda$ = change in wavelength, v = recession velocity.

$$v = H_0 d$$

Where v = recession velocity (km s^{-1} or m s^{-1}), H_0 = Hubble constant ($\approx 67\text{--}73 \text{ km s}^{-1} \text{ Mpc}^{-1}$), d = distance to the galaxy (Mpc or m).

- Hubble's law shows that more distant galaxies are receding faster, supporting the Big Bang theory
- The Hubble constant gives an estimate of the age of the universe: $t \approx 1/H_0$

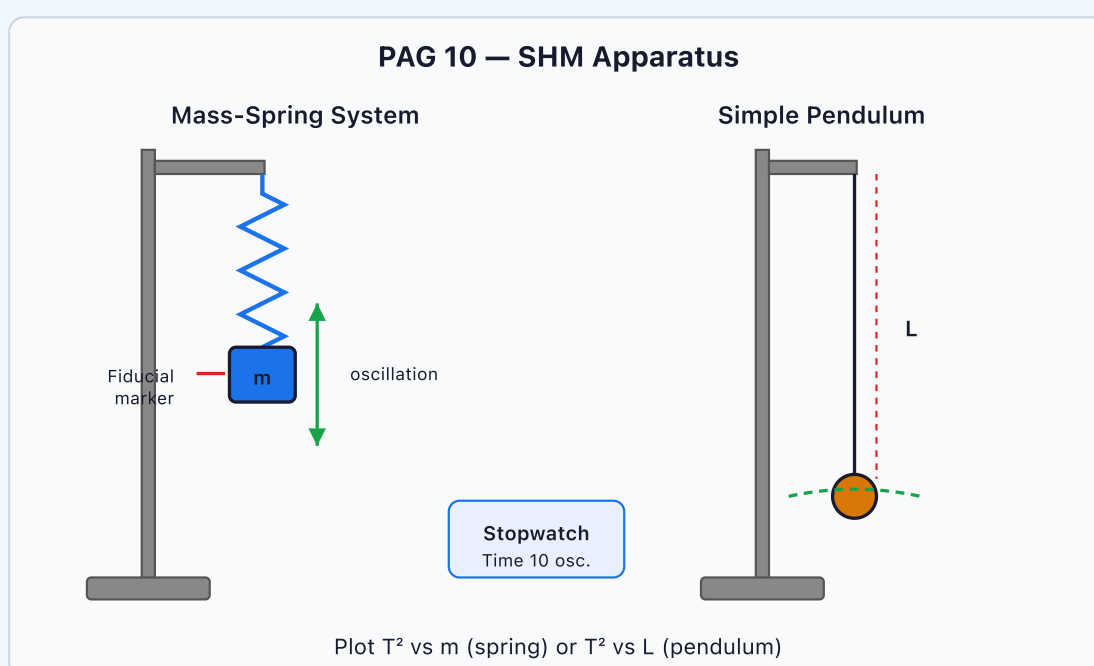
Evidence for the Big Bang

1. **Redshift of galaxies:** All distant galaxies show redshift, indicating the universe is expanding. Extrapolating backwards implies all matter was once concentrated at a single point.
2. **Cosmic Microwave Background (CMB) radiation:** Uniform microwave radiation detected from all directions, with a blackbody spectrum corresponding to $\sim 2.7 \text{ K}$. This is the cooled remnant of the hot, dense early universe, predicted by the Big Bang model.

3. **Abundance of light elements:** The observed ratios of hydrogen (~75%) and helium (~25%) match predictions from Big Bang nucleosynthesis.

Exam Tip: The CMB is not the same as starlight. It is thermal radiation from the early universe, now redshifted to microwave wavelengths due to the expansion of space. Its near-perfect uniformity and blackbody spectrum are strong evidence for the Big Bang.

PAG 10: Investigation of SHM



Apparatus

- **Mass-spring:** Helical spring, clamp stand, mass hanger + slotted masses, stopwatch, ruler, fiducial marker
- **Pendulum:** String, small dense bob, clamp stand, protractor, stopwatch, metre ruler

Method

1. **Mass-spring:** Attach a known mass m to the spring. Displace and release. Time 10 complete oscillations and divide by 10 to get T . Repeat 3 times and find the mean T . Repeat for at least 6 different masses.

2. **Pendulum:** Measure the length L from the pivot to the centre of the bob. Displace to a small angle (<10 degrees) and release. Time 10 complete oscillations. Repeat for at least 6 different lengths.
3. For mass-spring: plot T^2 against m . Gradient $= 4\pi^2/k$, so $k = 4\pi^2/\text{gradient}$.
4. For pendulum: plot T^2 against L . Gradient $= 4\pi^2/g$, so $g = 4\pi^2/\text{gradient}$.

Safety

- Use a retort stand with a heavy base to prevent toppling
- Place a cushion below the mass-spring in case the spring detaches
- Use small amplitudes to keep within the SHM approximation

Analysis

Mass-spring: $T = 2\pi\sqrt{m/k}$, so $T^2 = (4\pi^2/k)m$. Plot T^2 vs m for a straight line through the origin with gradient $= 4\pi^2/k$.

Pendulum: $T = 2\pi\sqrt{L/g}$, so $T^2 = (4\pi^2/g)L$. Plot T^2 vs L for a straight line through the origin with gradient $= 4\pi^2/g$.

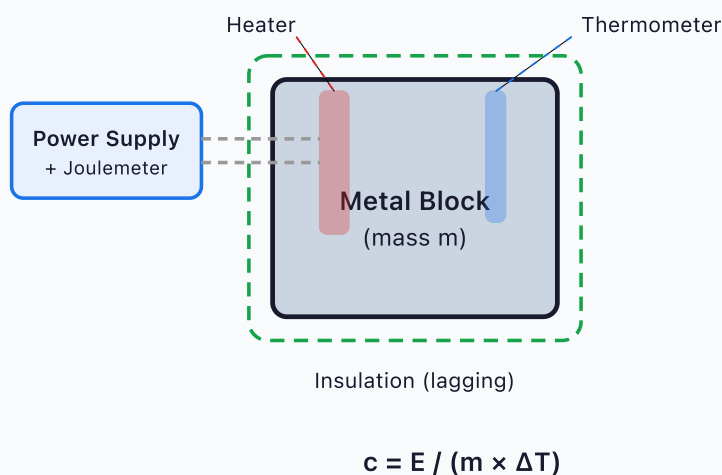
Sources of Error & Improvements

Source of Error	Type	Improvement
Human reaction time in starting/stopping the stopwatch	Random	Time 10+ oscillations and divide; use a fiducial marker at the equilibrium position; use light gates
Pendulum swinging in a circle instead of a plane	Systematic	Release carefully; use a short initial displacement
Measuring L to centre of bob is imprecise	Random	Use vernier calipers to measure bob diameter; $L = \text{string length} + \text{radius}$
Damping reduces amplitude during timing	Systematic	SHM period is independent of amplitude (if small), so this only affects accuracy if amplitude becomes too large

Exam Tip: The period of a simple pendulum is independent of mass and amplitude (for small angles). The period of a mass-spring system is independent of amplitude. Always verify these by checking that T remains constant when you vary the irrelevant variable.

PAG 11: Specific Heat Capacity

PAG 11 — Specific Heat Capacity Apparatus



Apparatus

- Metal block (aluminium or copper) with holes for heater and thermometer
- Electric immersion heater
- Thermometer (or temperature probe/datalogger)
- Joulemeter (or ammeter + voltmeter + stopwatch to calculate $E = IVt$)
- Power supply
- Insulation (lagging) around the block
- Digital balance to measure mass of block

Method

1. Measure and record the mass m of the metal block using a balance
2. Insert the heater and thermometer into the holes; add a drop of oil for good thermal contact
3. Wrap the block in insulation to reduce heat loss
4. Record the initial temperature θ_1
5. Turn on the heater and record the energy E supplied (from the joulemeter, or calculate $E = IVt$)
6. When the temperature has risen by $\sim 10\text{--}20\text{ K}$, switch off the heater and record the maximum temperature θ_2
7. Calculate $c = E / (m \times \Delta\theta)$ where $\Delta\theta = \theta_2 - \theta_1$

Safety

- The metal block gets very hot — use a heatproof mat and handle with tongs/gloves
- Ensure the heater is fully inserted into the block before switching on
- Do not leave the apparatus unattended while heating

Analysis

From $Q = mc\Delta\theta$, rearranging gives $c = Q/(m\Delta\theta)$. If using a datalogger, plot θ against t and use the gradient with $P = mc(d\theta/dt)$ to find c . Alternatively, plot E against $\Delta\theta$ — the gradient $= mc$, so $c = \text{gradient}/m$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic (gives c too high)	Insulate the block well; heat and cool by the same amount above and below room temperature
Block temperature is not uniform	Systematic	Wait for thermal equilibrium; use a well-lagged block; stir (if liquid)
Energy lost in wires and connections	Systematic	Place voltmeter across heater (not power supply) to measure actual energy input

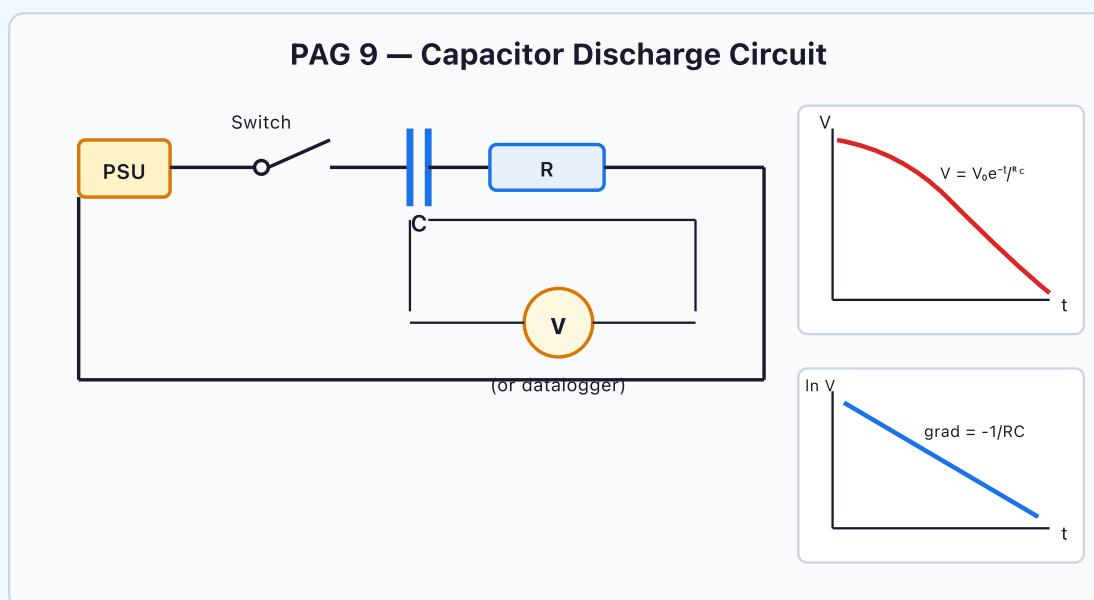
Temperature continues rising after heater switched off

Systematic

Record the maximum temperature reached after switching off (thermal lag)

Exam Tip: The experimental value of c is usually higher than the accepted value because some energy is lost to the surroundings. Better insulation reduces this systematic error.

PAG 9: Capacitor Charge and Discharge



Apparatus

- Capacitor (e.g. 470 μF or 1000 μF electrolytic)
- Resistor (e.g. 10 $\text{k}\Omega$ or 100 $\text{k}\Omega$ — chosen so RC gives a reasonable time constant)
- DC power supply or battery
- Two-way switch (or two separate switches for charge/discharge)
- Voltmeter or datalogger with voltage sensor across the capacitor
- Stopwatch (if no datalogger)

Method

1. Connect the circuit: power supply, switch, capacitor, and resistor. Voltmeter (or datalogger) across the capacitor.
2. Close the switch to charge the capacitor fully (wait for V to reach V_0)
3. Switch to discharge through the resistor. Start the stopwatch/datalogger.
4. Record V at regular time intervals (e.g. every 10 s) until V is negligible
5. Plot V against t to show exponential decay
6. Plot $\ln(V)$ against t — this should give a straight line with gradient $= -1/RC$
7. From the gradient, calculate C (if R is known) or RC (the time constant)

Safety

- Ensure correct polarity for electrolytic capacitors
- Do not exceed the rated voltage of the capacitor
- Discharge capacitors before handling

Analysis

From $V = V_0 e^{-t/RC}$, taking natural logs: $\ln(V) = \ln(V_0) - t/RC$. Plotting $\ln(V)$ vs t gives a straight line with gradient $= -1/RC$ and y -intercept $= \ln(V_0)$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Voltmeter has finite resistance, providing an additional discharge path	Systematic	Use a high-resistance voltmeter or datalogger with very high input impedance
Timing errors when reading V manually	Random	Use a datalogger for continuous automated recording
Capacitor value tolerance (e.g. $\pm 20\%$)	Systematic	Use the experimental value from the graph rather than the labelled value

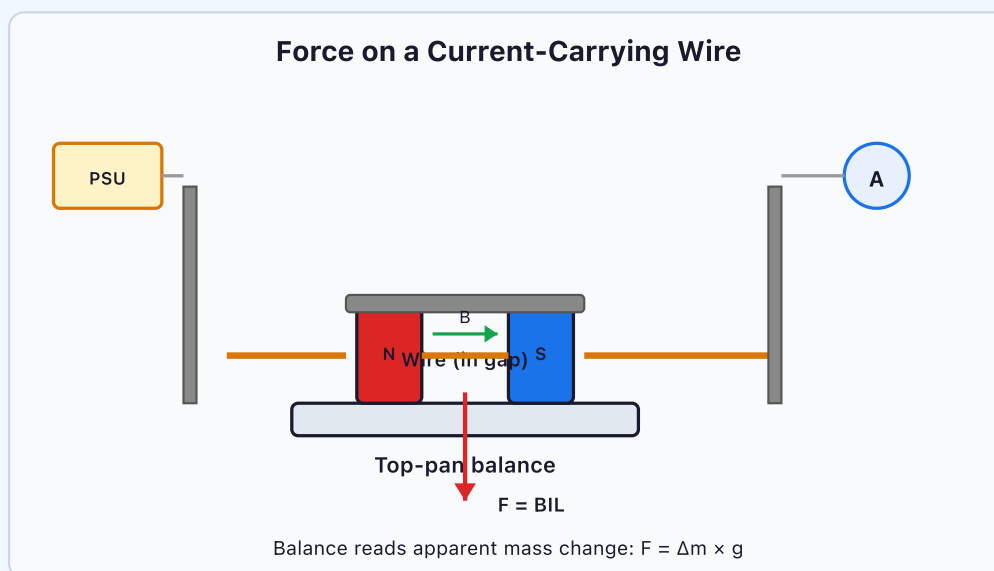
Temperature changes affecting R

Systematic

Use a high-stability resistor; perform experiment quickly

Exam Tip: The time constant $\tau = RC$ is the time for V to fall to V_0/e (about 37%). On the $\ln(V)$ vs t graph, $1/\tau$ is the magnitude of the gradient. Choose R and C so that τ is between 10 s and 100 s for practical data collection.

🔧 Practical (not a PAG): Force on a Current-Carrying Wire



Apparatus

- Strong magnets (horseshoe or U-shaped) mounted on a yoke, placed on a top-pan balance
- Rigid straight wire passing through the magnetic field gap, supported by clamp stands
- Variable DC power supply and ammeter in series with the wire
- Top-pan balance (resolution 0.01 g or better)
- Ruler to measure length L of wire in the field

Method

1. Set up the magnets on the top-pan balance with the wire passing horizontally through the gap between the poles
2. Zero the balance with no current flowing
3. Turn on the current and record the ammeter reading I and the new balance reading
4. The change in mass reading Δm gives the force: $F = \Delta m \times g$ (by Newton's Third Law, the force on the wire equals the force on the magnets)
5. Vary the current in at least 6 steps and record F and I at each
6. Plot F (y-axis) against I (x-axis)
7. Gradient = BL , so $B = \text{gradient} / L$

Safety

- Keep strong magnets away from electronic devices and pacemakers
- Do not exceed the current rating of the wire
- Switch off between readings to prevent wire heating

Analysis

From $F = BIL$, with B and L constant, F is proportional to I . The gradient of F vs I gives BL . Measure L with a ruler to calculate $B = \text{gradient}/L$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Magnetic field is not perfectly uniform across the gap	Systematic	Use a narrower gap or shaped pole pieces for a more uniform field
Wire not perfectly perpendicular to the field	Systematic	Align carefully using a set square; ensure wire is horizontal
Balance fluctuates due to vibrations	Random	Place on an anti-vibration surface; take readings when steady

Difficulty measuring exact length of wire in the field

Random

Use sharp-edged pole pieces so the field boundary is well-defined

Exam Tip: By Newton's Third Law, the force on the wire (upward) is equal and opposite to the force on the magnets (downward). The balance reads an increased mass when the force pushes the magnets down, and decreased mass when it pushes them up.

Worked Example: A satellite orbits Earth at an altitude of 400 km. Calculate its orbital speed and period. ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m, $G = 6.674 \times 10^{-11}$ N m² kg⁻²)

$$r = R_E + h = 6.37 \times 10^6 + 400 \times 10^3 = 6.77 \times 10^6 \text{ m}$$

$$v = \sqrt{GM/r} = \sqrt{(6.674 \times 10^{-11} \times 5.97 \times 10^{24}) / 6.77 \times 10^6} = \sqrt{5.89 \times 10^7} = \mathbf{7670 \text{ m s}^{-1} \approx 7.7 \text{ km s}^{-1}}$$

$$T = 2\pi r / v = 2\pi \times 6.77 \times 10^6 / 7670 = \mathbf{5540 \text{ s} \approx 92 \text{ minutes}}$$

This is the orbit of the International Space Station, which completes about 16 orbits per day.

Worked Example: Calculate the rms speed of helium atoms at 300 K. ($M = 4.0 \times 10^{-3}$ kg mol⁻¹, $R = 8.314$ J mol⁻¹ K⁻¹)

$$c_{\text{rms}} = \sqrt{3RT/M} = \sqrt{(3 \times 8.314 \times 300 / 4.0 \times 10^{-3})} = \sqrt{1.87 \times 10^6} = \mathbf{1370 \text{ m s}^{-1}}$$

Helium atoms move about 2.6 times faster than nitrogen molecules at the same temperature because they are 7 times lighter ($c_{\text{rms}} \propto 1/\sqrt{M}$).